R PROGRAMMING

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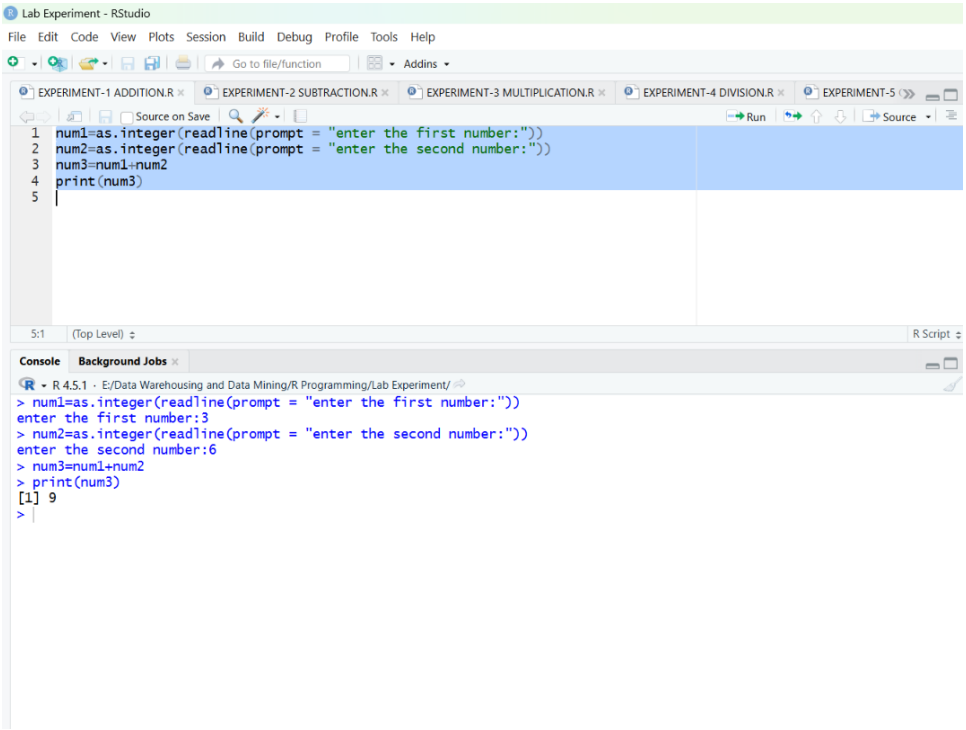
EXPERIMENT-1
ADDITION:

AIM:
To prove the program for addition using R-tool.

PROGRAM:

```
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1+num2
print(num3)
```

OUTPUT:



RESULT:
Thus the basic program addition is executed successfully.

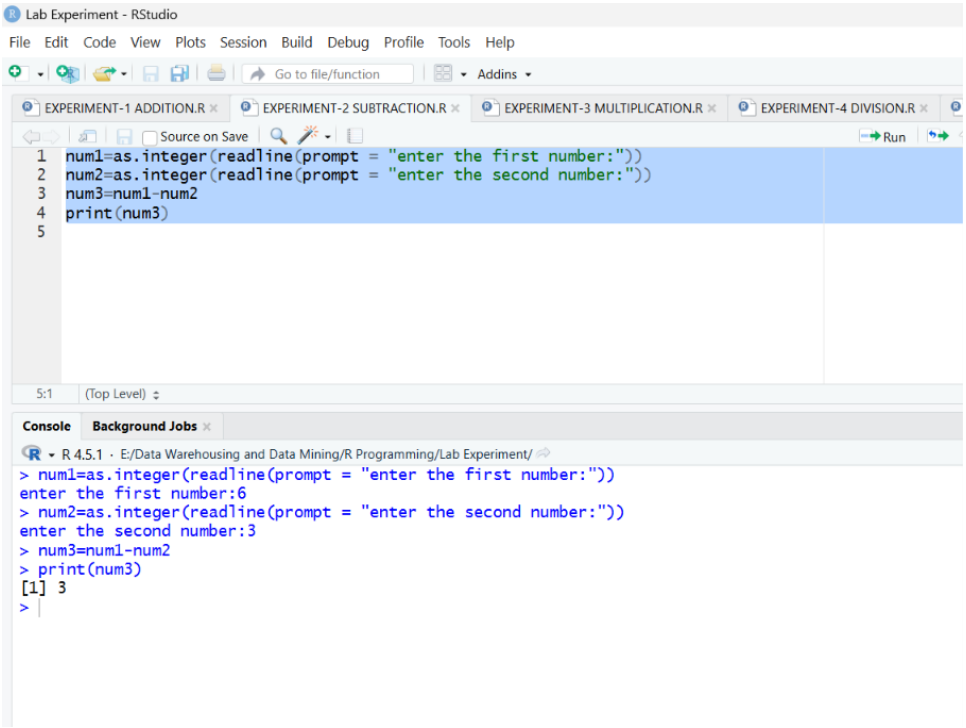
EXPERIMENT-2
SUBTRACTION:

AIM:
To prove the program for subtraction using R-tool.

PROGRAM:

```
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1-num2
print(num3)
```

OUTPUT:



RESULT:
Thus, the basic program subtraction is executed successfully.

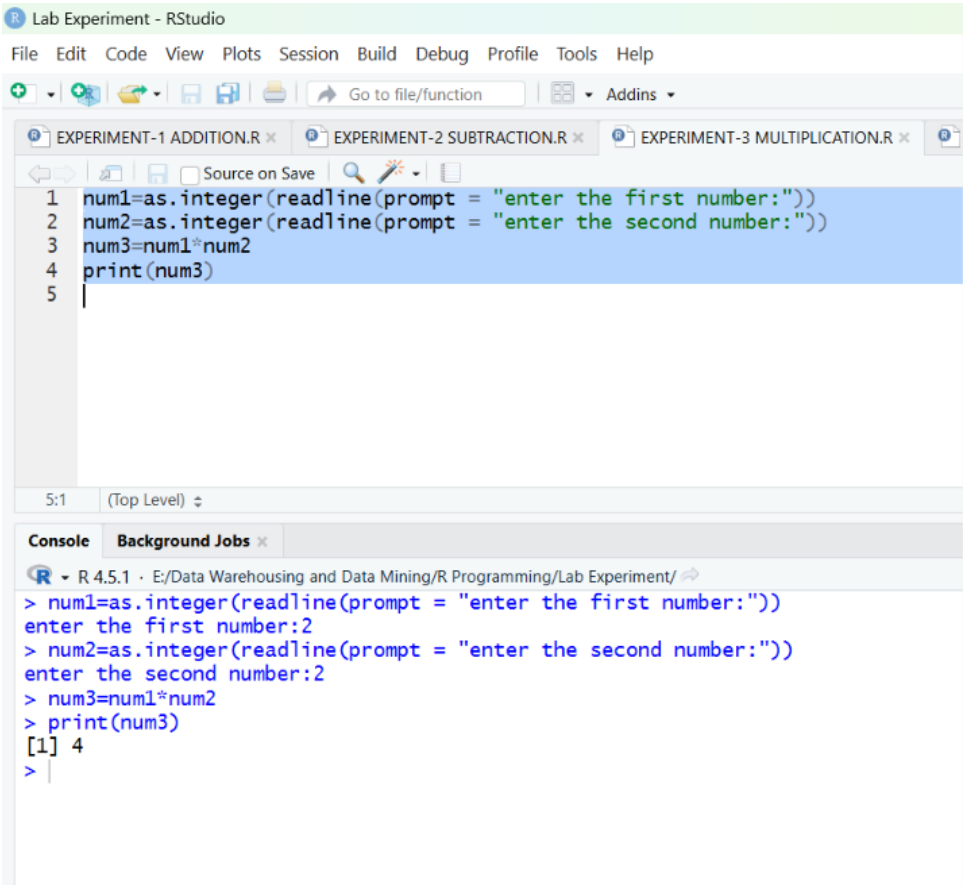
EXPERIMENT-3
MULTIPLICATION:

AIM:
To prove the program for multiplication using R-tool.

PROGRAM:

```
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1*num2
print(num3)
```

OUTPUT:



RESULT:
Thus, the basic program multiplication is executed successfully.

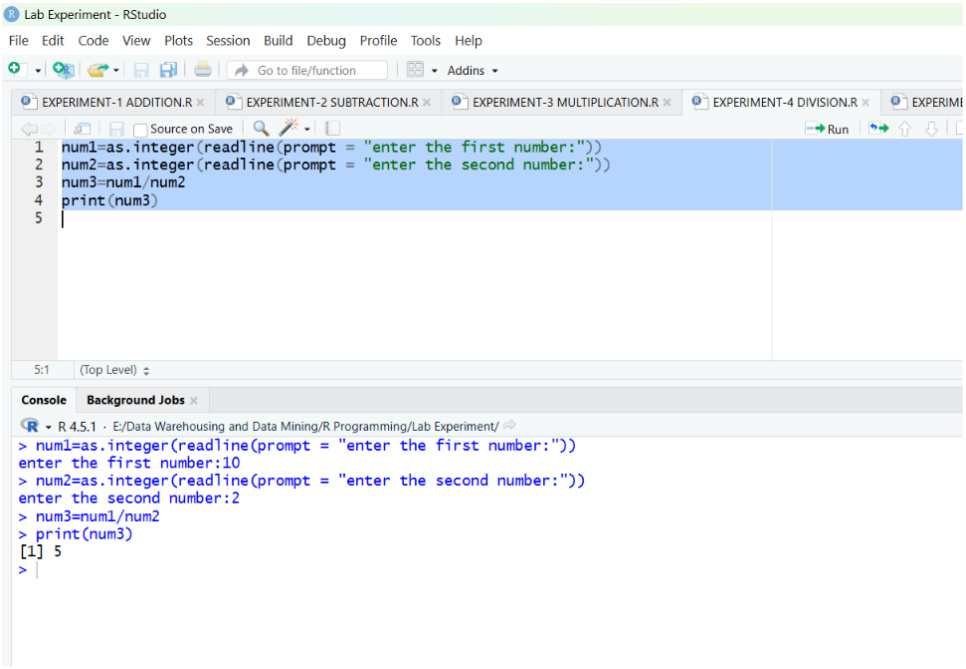
EXPERIMENT-4
DIVISION:

AIM:
To prove the program for division using R-tool.

PROGRAM:

```
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1/num2
print(num3)
```

OUTPUT:



RESULT:
Thus, the basic program division was executed successfully.

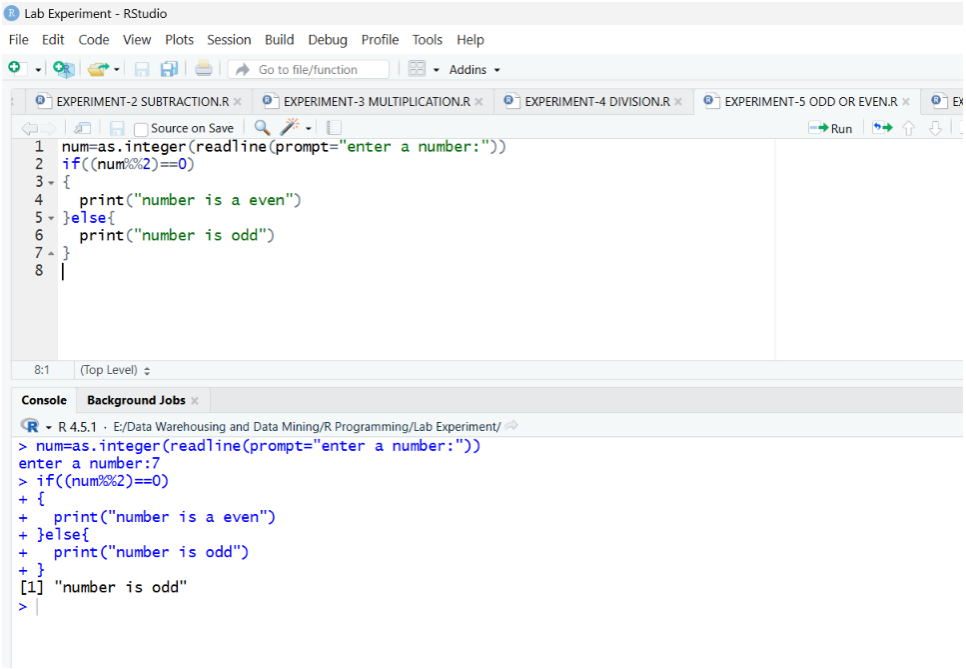
EXPERIMENT-5
ODD OR EVEN:

AIM:
To write the program for odd or even using R-tool.

PROGRAM:

```
num=as.integer(readline(prompt="enter a number:"))
if((num%%2)==0)
{
  print("number is a even")
}else{
  print("number is odd")
}
```

OUTPUT:



RESULT:
Thus, the basic program odd or even was executed successfully.

EXPERIMENT-6
MEAN,MEDIAN,MODE:

AIM:
To write the program for mean,median,mode.

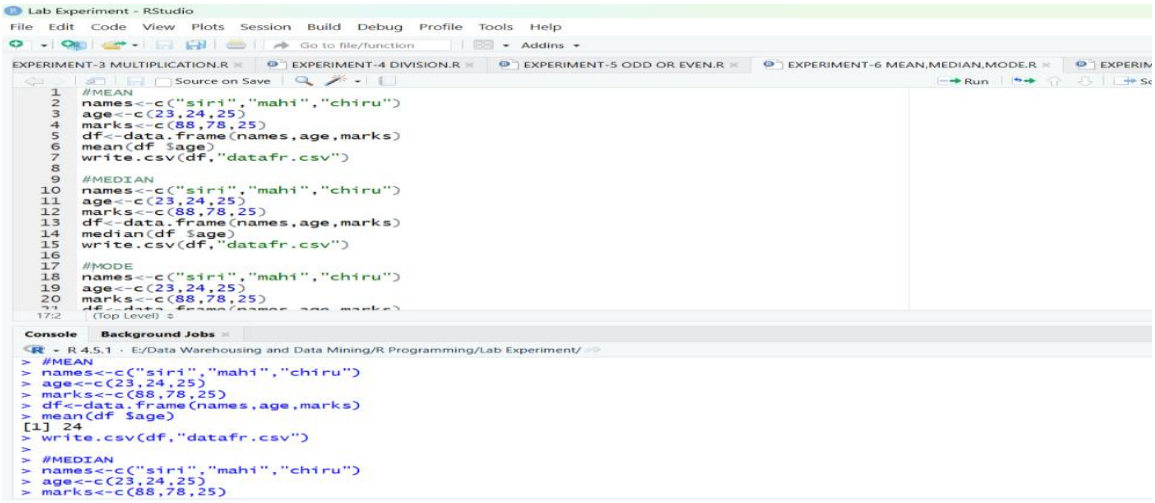
PROGRAM:

```
#MEAN
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
mean(df $age)
write.csv(df,"datafr.csv")

#MEDIAN
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
median(df $age)
write.csv(df,"datafr.csv")

#MODE
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
mode(df $age)
write.csv(df,"datafr.csv")
```

OUTPUT:



RESULT:
Thus the central tendency and measure of dispersion is executed successfully.

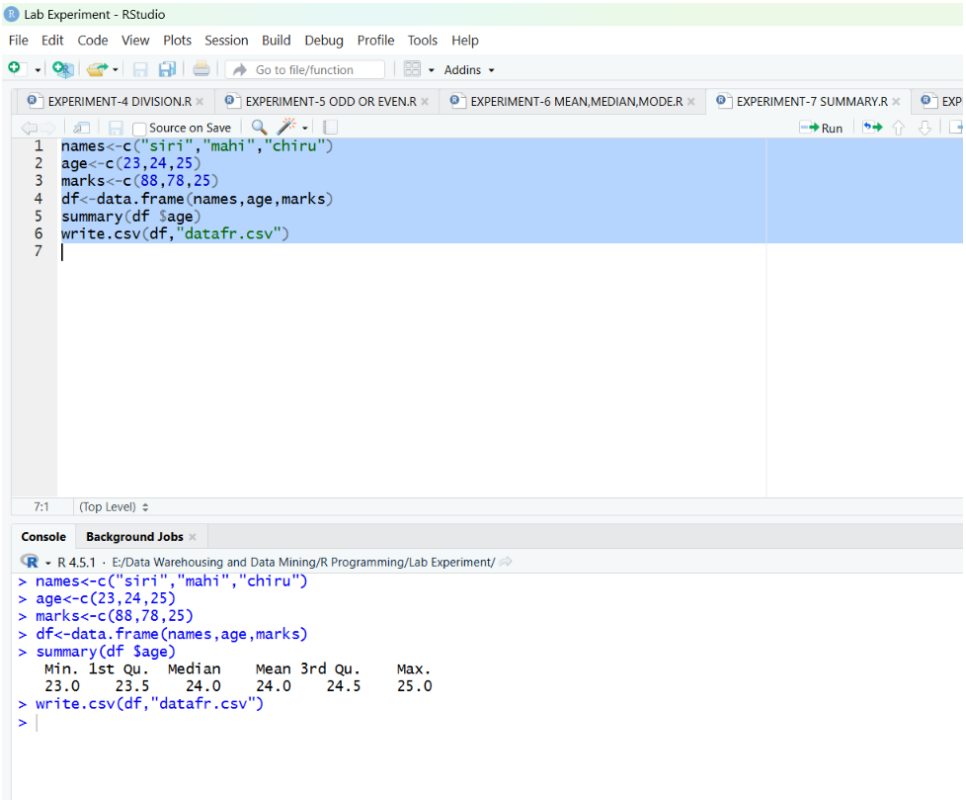
EXPERIMENT-7
SUMMARY:

AIM:
To write the program for summary using R-tool.

PROGRAM:

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
summary(df $age)
write.csv(df,"datafr.csv")
```

OUTPUT:



RESULT:
Thus, the central tendency and measure of dispersion is executed successfully.

EXPERIMENT-8
GREATER AMONG THREE NUMBERS:

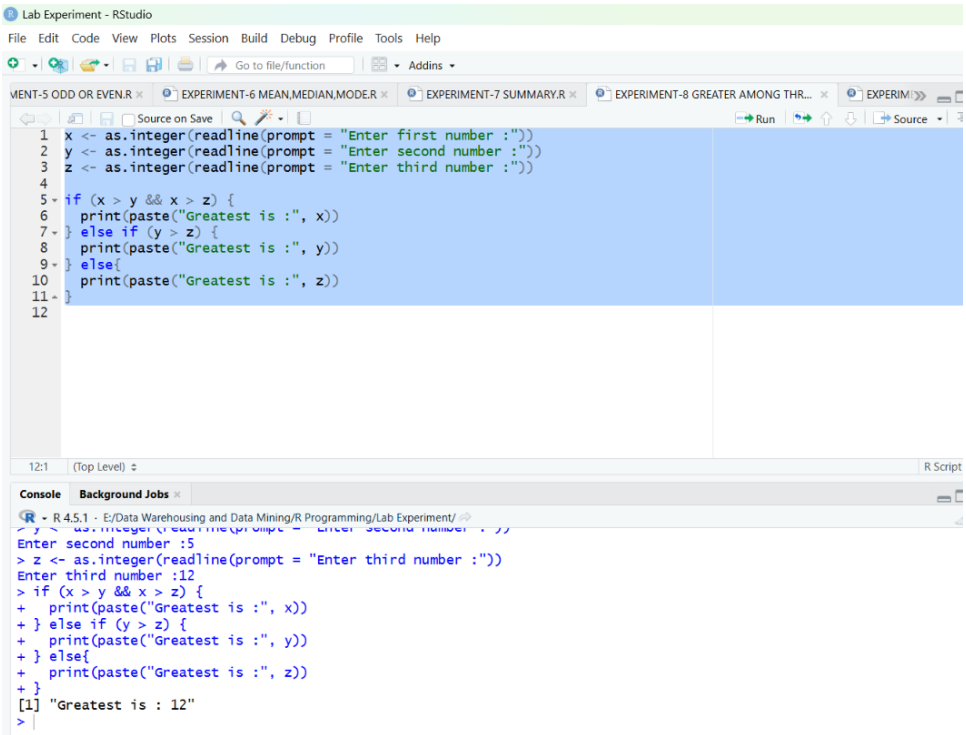
AIM:
To write the program for the greatest among three numbers.

PROGRAM:

```
x <- as.integer(readline(prompt = "Enter first number :"))
y <- as.integer(readline(prompt = "Enter second number :"))
z <- as.integer(readline(prompt = "Enter third number :"))


if (x > y && x > z) {
  print(paste("Greatest is :", x))
} else if (y > z) {
  print(paste("Greatest is :", y))
} else{
  print(paste("Greatest is :", z))
}
```

OUTPUT:



RESULT:
Thus, the greatest among the three numbers was executed successfully.

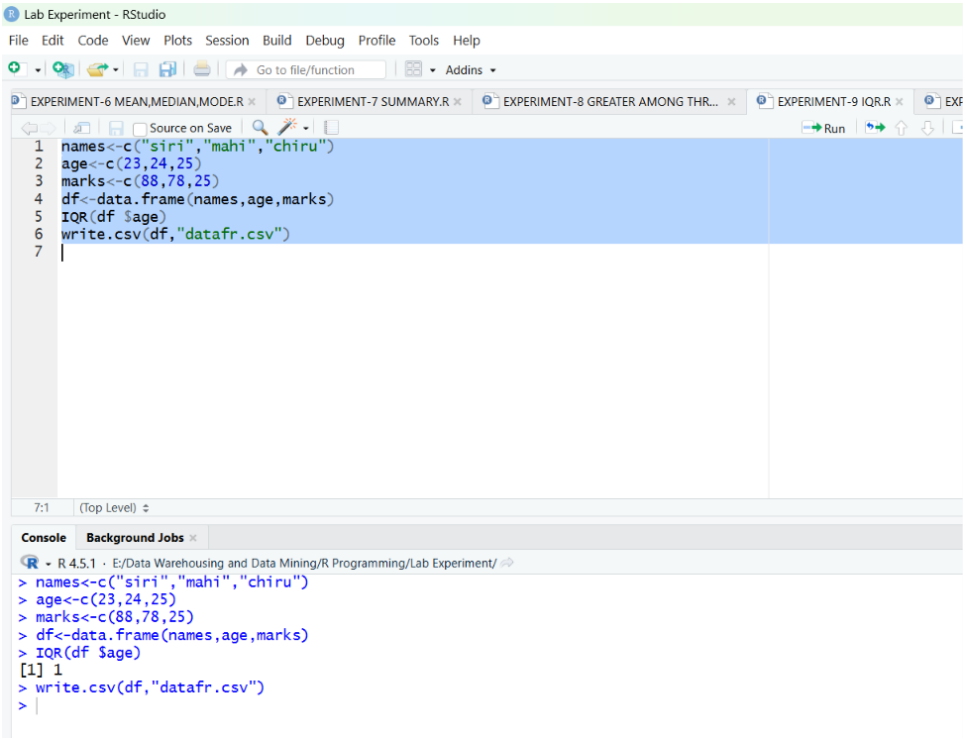
EXPERIMENT-9
IQR:

AIM:
To write the program for central tendency and data dispersion measures using R tool.

PROGRAM:

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
IQR(df $age)
write.csv(df,"datafr.csv")
```

OUTPUT:



RESULT:
Thus, the program for central tendency and data dispersion measures was executed successfully.

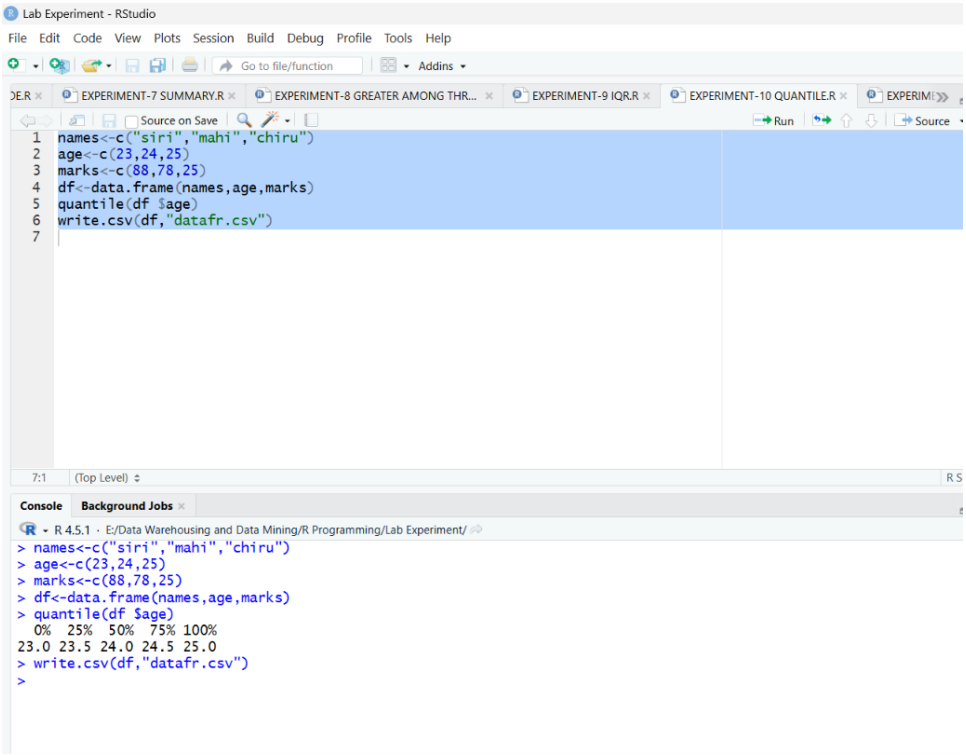
EXPERIMENT-10
QUANTILE:

AIM:
To write the program for central tendency and data dispersion measures.

PROGRAM:

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
quantile(df $age)
write.csv(df,"datafr.csv")
```

OUTPUT:



RESULT:
Thus, the program for central tendency and data dispersion measures was executed successfully.

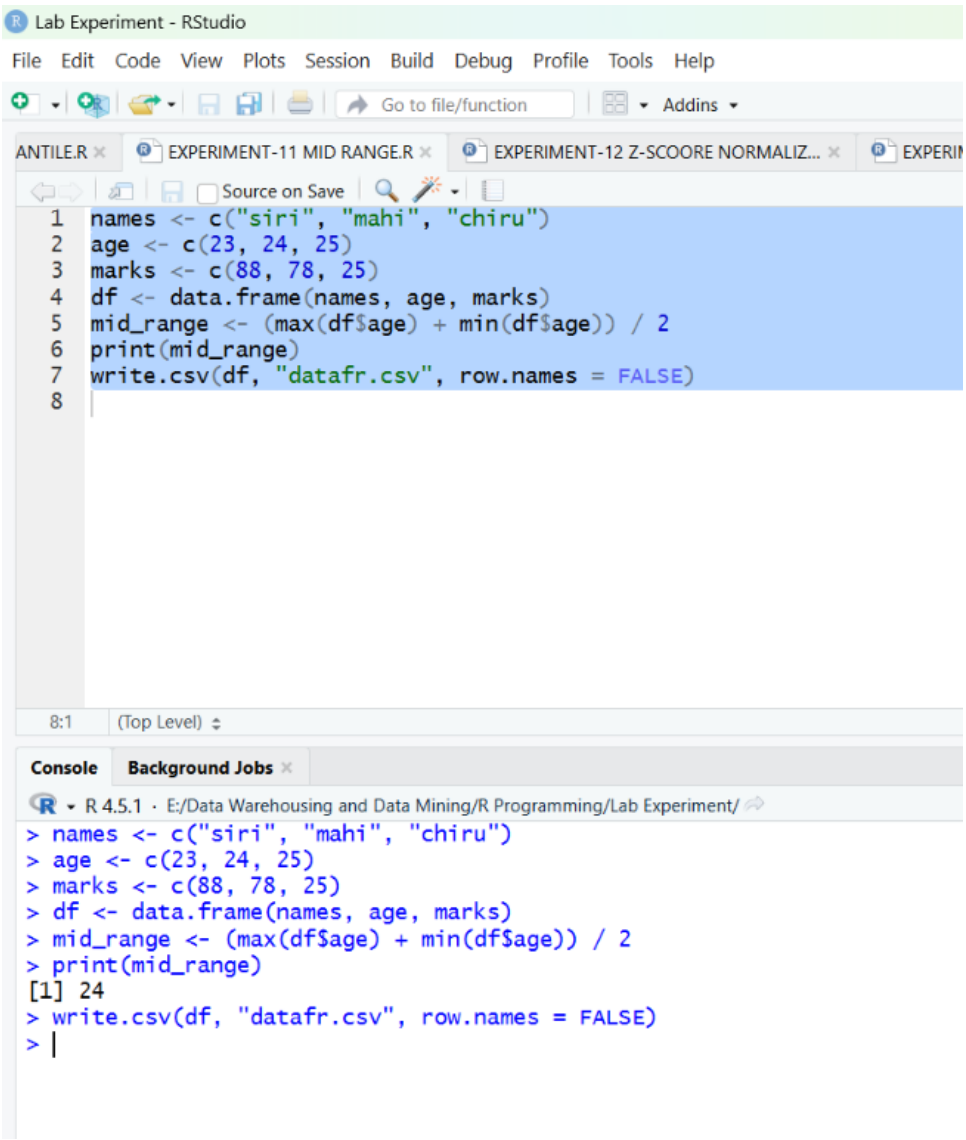
EXPERIMENT-11
MID RANGE:

AIM:
To write the program for central tendency and data dispersion measures.

PROGRAM:

```
names <- c("siri", "mahi", "chiru")
age <- c(23, 24, 25)
marks <- c(88, 78, 25)
df <- data.frame(names, age, marks)
mid_range <- (max(df$age) + min(df$age)) / 2
print(mid_range)
write.csv(df, "datafr.csv", row.names = FALSE)
```

OUTPUT:



RESULT:
Thus, the program for central tendency and data dispersion measures was executed successfully.

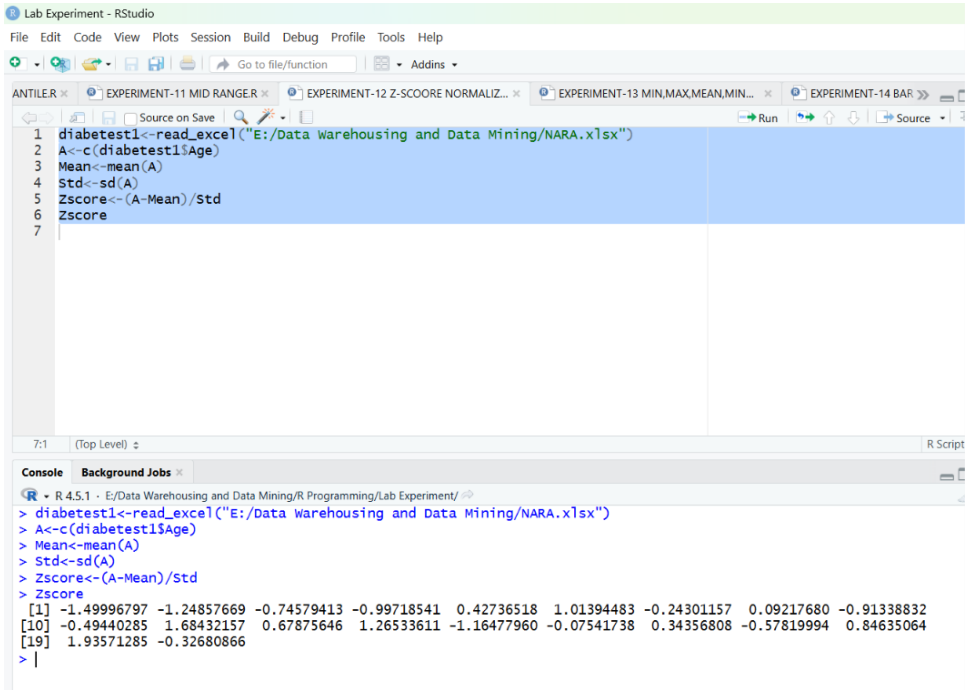
EXPERIMENT-12
Z-SCORE NORMALIZATION:

AIM:
To write the program for Z-score normalization using R-tool.

PROGRAM:

```
diabetest1<-read_excel("E:/Data Warehousing and Data Mining/NARA.xlsx")
A<-c(diabetest1$Age)
Mean<-mean(A)
Std<-sd(A)
Zscore<-(A-Mean)/Std
Zscore
```

OUTPUT:



RESULT:
Thus, the Z-score normalization using R tool was executed successfully.

EXPERIMENT-13
MIN,MAX,MEAN,MINMAX:

AIM:
To write the program for minimum, maximum, mean and min-max normalization using R-tool.

PROGRAM:

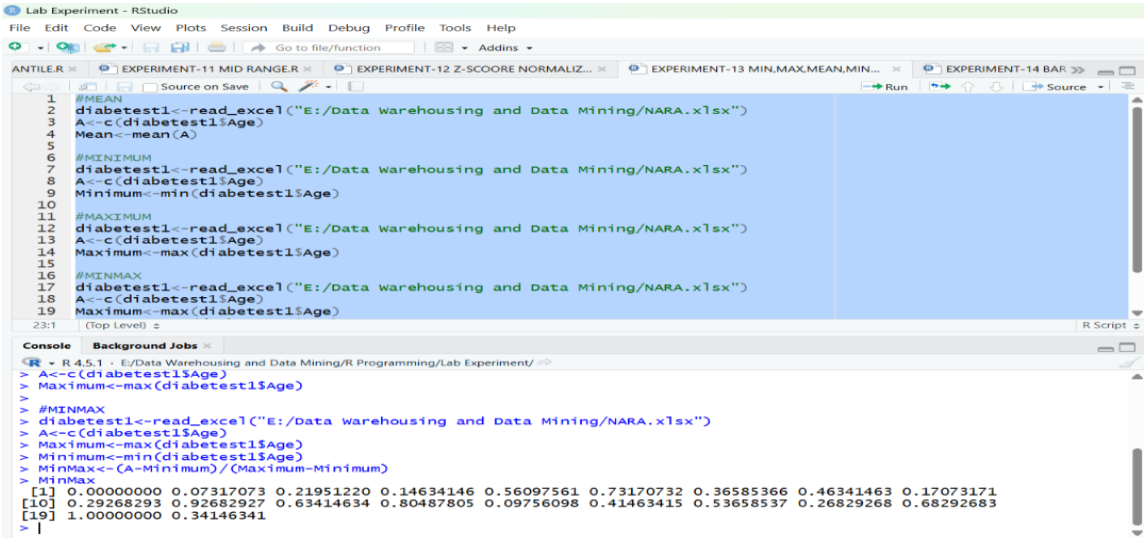
```
#MEAN
diabetest1<-read_excel("E:/Data Warehousing and Data Mining/NARA.xlsx")
A<-c(diabetest1$Age)
Mean<-mean(A)

#MINIMUM
diabetest1<-read_excel("E:/Data Warehousing and Data Mining/NARA.xlsx")
A<-c(diabetest1$Age)
Minimum<-min(diabetest1$Age)

#MAXIMUM
diabetest1<-read_excel("E:/Data Warehousing and Data Mining/NARA.xlsx")
A<-c(diabetest1$Age)
Maximum<-max(diabetest1$Age)

#MINMAX
diabetest1<-read_excel("E:/Data Warehousing and Data Mining/NARA.xlsx")
A<-c(diabetest1$Age)
Maximum<-max(diabetest1$Age)
Minimum<-min(diabetest1$Age)
MinMax<-(A-Minimum)/(Maximum-Minimum)
MinMax
```

OUTPUT:



RESULT:
Thus, the program for minimum, maximum, mean and min-max normalization was executed successfully.

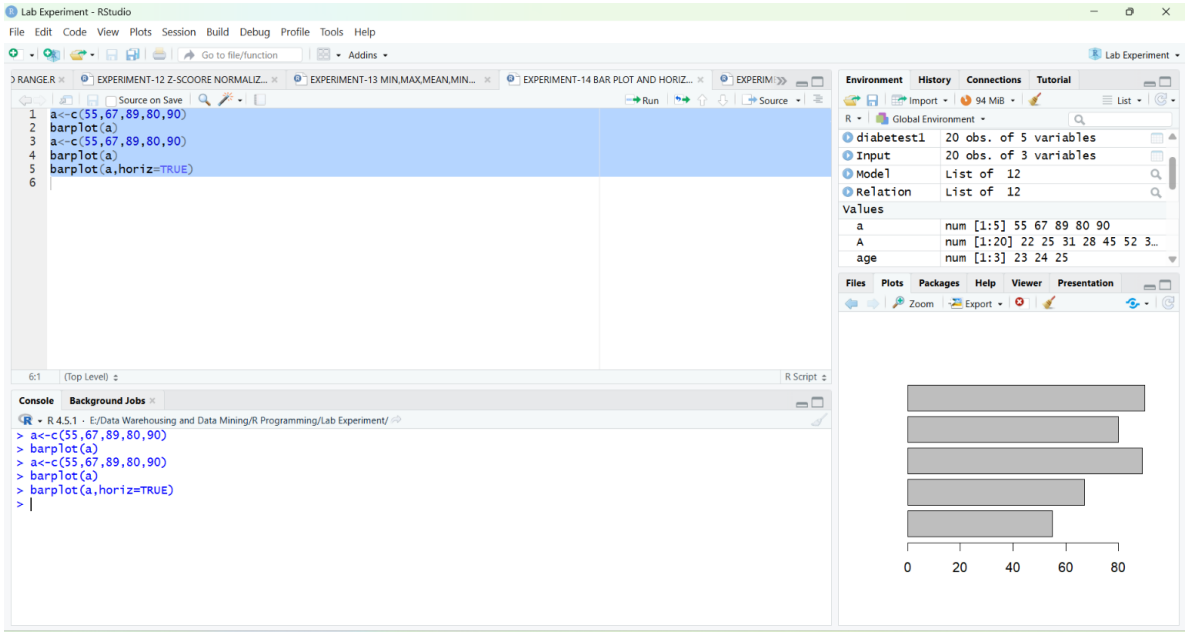
EXPERIMENT-14
BAR PLOT AND HORIZONTAL BAR PLOT:

AIM:
To write the program for bar plot and horizontal bar plot using R-tool.

PROGRAM:

```
a<-c(55,67,89,80,90)
barplot(a)
a<-c(55,67,89,80,90)
barplot(a)
barplot(a,horiz=TRUE)
```

OUTPUT:



RESULT:
Thus, the bar plot and horizontal bar plot were executed successfully.

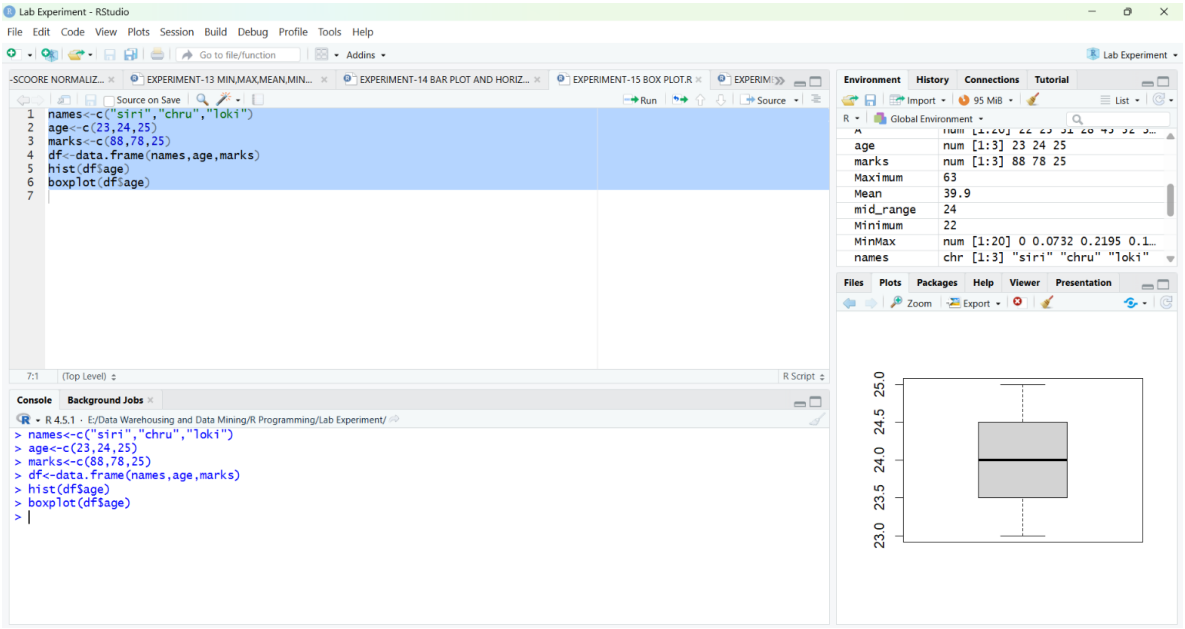
EXPERIMENT-15
BOX PLOT:

AIM:
To write the program for box plot using R-tool.

PROGRAM:

```
names<-c("siri","chru","loki")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
hist(df$age)
boxplot(df$age)
```

OUTPUT:



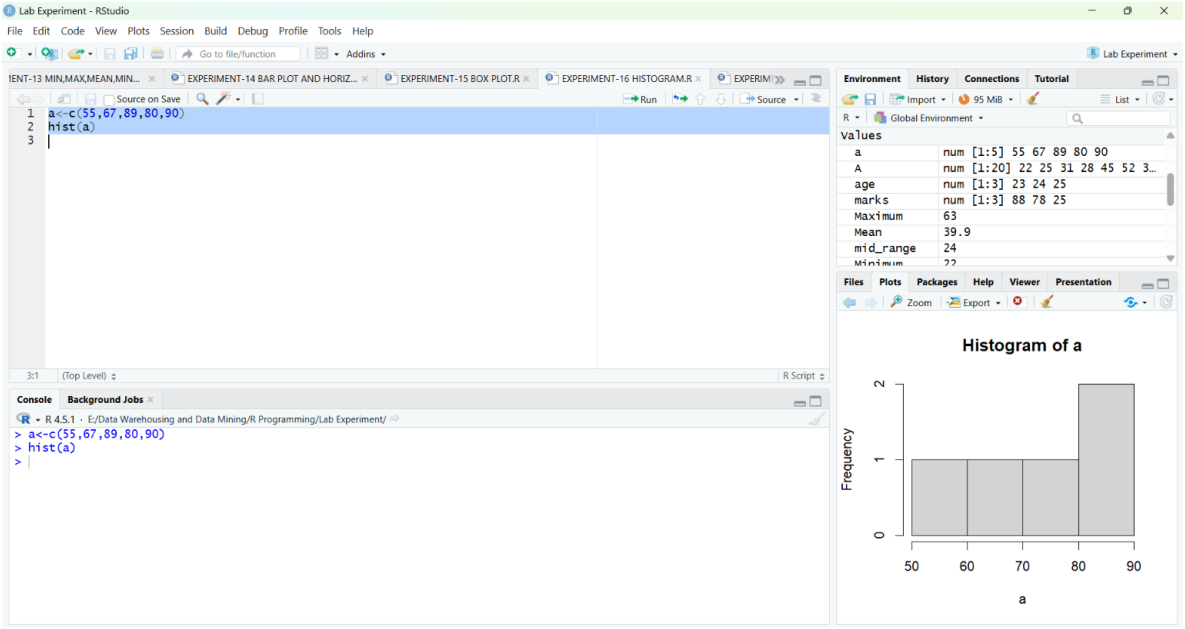
RESULT:
Thus, the box plot was executed successfully.

EXPERIMENT-16
HISTOGRAM:

AIM:
To write the program for histogram using R-tool.

PROGRAM:
a<-c(55,67,89,80,90)
hist(a)

OUTPUT:



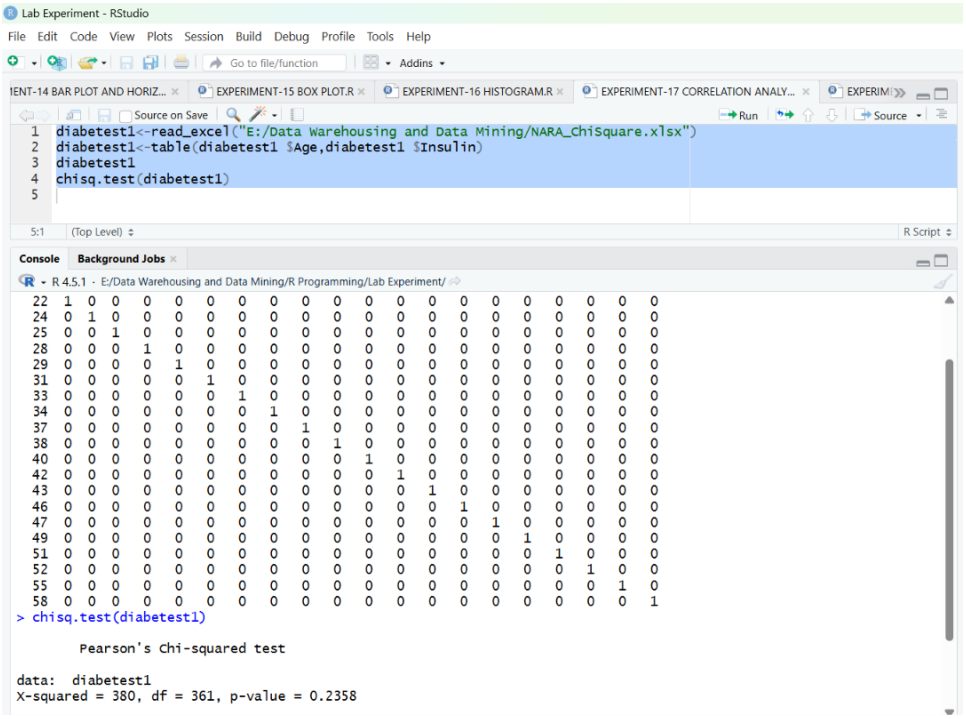
RESULT:
Thus, the histogram was executed successfully.

EXPERIMENT-17
CORRELATION ANALYSIS:

AIM:
To write the program for correlation analysis using the chi-square test in R-tool.

PROGRAM:
```
diabetest1<-read_excel("E:/Data Warehousing and Data Mining/NARA_ChiSquare.xlsx")
diabetest1<-table(diabetest1 $Age,diabetest1 $Insulin)
diabetest1
chisq.test(diabetest1)
```

OUTPUT:



RESULT:
Thus, the correlation analysis using the chi-square test was executed successfully.

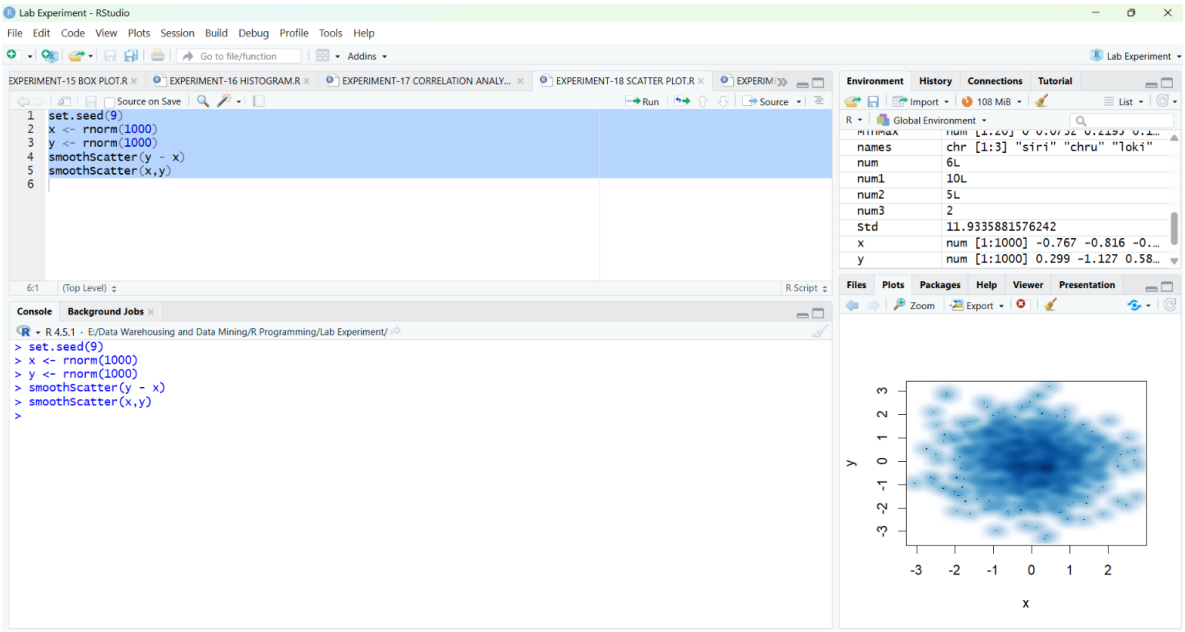
EXPERIMENT-18
SCATTER PLOT:

AIM:
To write the program for scatter plot using R-tool.

PROGRAM:

```
set.seed(9)
x <- rnorm(1000)
y <- rnorm(1000)
smoothScatter(y - x)
smoothScatter(x,y)
```

OUTPUT:



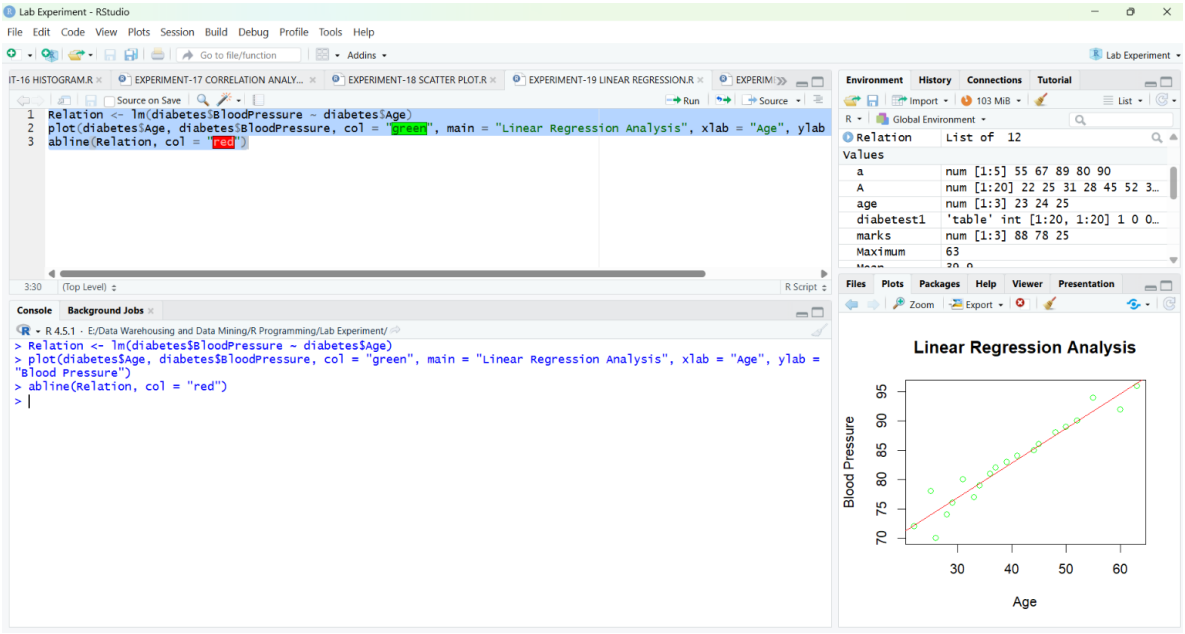
RESULT:
Thus, the scatter plot was executed successfully.

EXPERIMENT-19
LINEAR REGRESSION:

AIM:
To write the program for linear regression using R-tool.

PROGRAM:
Relation <- lm(diabetes$BloodPressure ~ diabetes$Age)
plot(diabetes$Age, diabetes$BloodPressure, col = "green", main = "Linear Regression Analysis", xlab = "Age", ylab = "Blood Pressure")
abline(Relation, col = "red")

OUTPUT:



RESULT:
Thus, the linear regression was executed successfully.

EXPERIMENT-20
MULTIPLE REGRESSION:

AIM:

To write the program for the multiple regression.

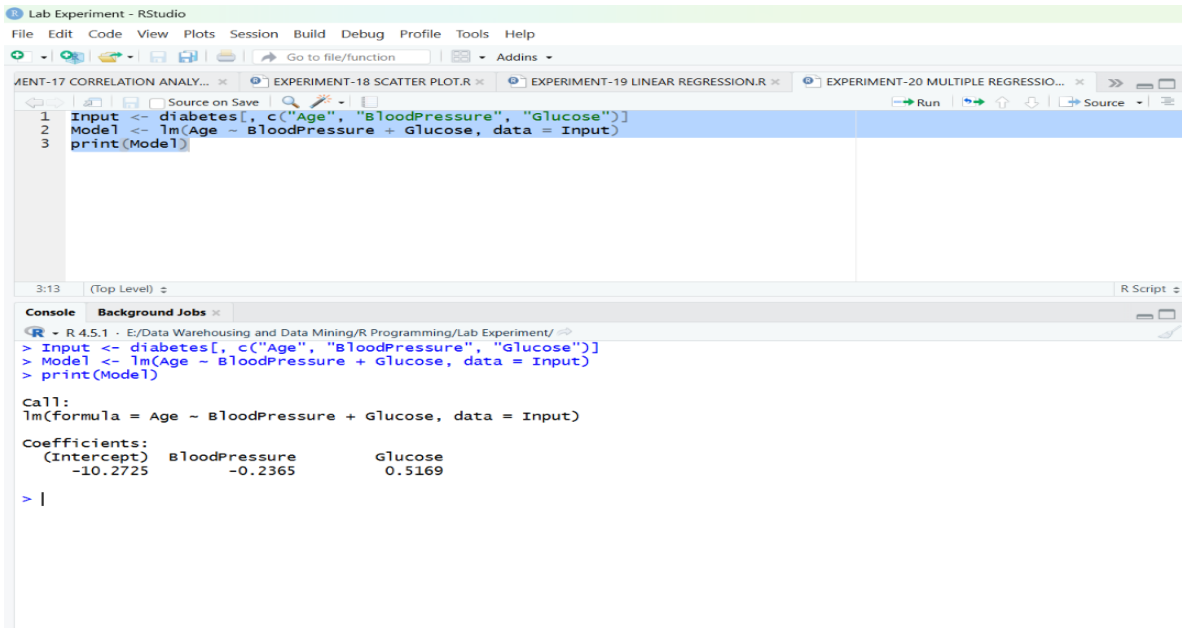
PROGRAM:

```
Input <- diabetes[, c("Age", "BloodPressure", "Glucose")]

Model <- lm(Age ~ BloodPressure + Glucose, data = Input)

print(Model)
```

OUTPUT:



RESULT:
Thus, the multiple regression was executed successfully.